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AI School Iteration 1.7

Further Preprocessing & ML Pipeline

<h4> By BOULGHERAIF Fatma Zohra </h4>

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01.Data Viz Review

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01. Iteration 1.6 Review: Data Viz

https://www.datacamp.com/tutorial/seaborn-python-tutorial

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02. Feature Scaling

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02. Feature Scaling: Why?

Problem: ML algorithms that use distances or gradients are sensitive to the **magnitude** of features.

Example:

- age: 25, 30, 45
- salary: 30,000 · 80,000 · 120,000

→ Salary **dominates the model.** Not because it matters more, but because it's bigger.

NB:
Affected: KNN · SVM · Linear/Logistic Regression · Neural Networks · PCA
Not affected: Decision Trees · Random Forests (split-based, not distance-based)

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02. Scaling: Absolute Maximum / Divide by Max

Formula: X' = X / X_max
Result: Values scaled to range [0, 1], but only if data has no negatives.

Example:

Original	**30**	**60**	**120**
Scaled	**0.25**	**0.5**	**1**

Limitation: Sensitive to outliers.
When to use: Quick normalization when you know the data is positive and clean.

02. Scaling: Min-Max Scaling

Formula: $X' = (X - X_{min}) / (X_{max} - X_{min})$
Result: Values always in range [0, 1]

Example:

Original	30	60	120
Scaled	0	0.33	1

Limitation: Sensitive to outliers.
When to use: When you know the bounds of a feature

02. Scaling: Standard Scaling / Z-score Normalization

Formula: X' = (X - μ) / σ
Result: Mean = 0, Std Dev = 1. No fixed range.

Example:

Original	**10**	**55**	**100**
Scaled	**-1.22**	**0**	**1.22**

Works well when data is roughly Gaussian. Robust for most ML algorithms

02. Scaling: Robust Scaler

Formula: (X - median) / IQR, aka (X - Q2) / (Q3 - Q1)
Result: Centered on median, scaled by interquartile range. Outliers have minimal effect.

Example:

Original	10	55	500
Standard Scaler	−0.57	+0.23	3.81
Robust Scaler	−0.67	0.00	+7.22

It maintains the structure of data

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03. Encoding Categorical Features

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03. Encoding Categorical Variables: Why?



Machine learning algorithms **work with numbers**, they cannot process raw text labels.

Rule of thumb

- **2 categories → Binary Encoding**
- **Many categories, no order → One-Hot Encoding**
- **Ordered categories → Label Encoding**

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03. Binary Encoding



When to use: Binary categorical variable (only 2 possible values)

Example: Gender

- Male: 0
- Female: 1

```
df['gender'] = (df['gender'] == 'Female').astype(int)
```

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03. One-Hot Encoding OHE

Color	is_Red	is_Blue	is_Green
Red	1	0	0
Blue	0	1	0
Green	0	0	1



When to use: Nominal categories with no natural order (e.g. color, city, product type)

Example: Color

- Red: [1, 0, 0]
- Green: [0, 1, 0]
- Blue: [0, 0, 1]

- *Red: [0, 0] . Green: [1, 0] . Blue: [0, 1]*

```
pd.get_dummies(df, columns=['color'], drop_first=True)
```

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03. Ordinal Encoding



When to use: Ordinal categories (where order matters)

Example: Salary

- Low: 0
- Medium: 1
- High: 2

Use sklearn.preprocessing.OrdinalEncoder, or simply, pandas df.replace method

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04. Missing Values

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04. Missing Values

- Remove rows: # of missing values < 5%
- Drop column: # of missing values > 50%
- Impute by mean/median: Use median when outliers are present; mean for clean distributions.
- Impute by mode: for categories

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05. Outliers

05. Outliers (Reminder)

Definition: Data points that sit far from the rest of the distribution.

Detection methods:

- IQR method: Outlier if $X < Q1 - 1.5 \times IQR$ or $X > Q3 + 1.5 \times IQR$
- Z-score: Outlier if $|z| > 3$

05. Outliers (Reminder)

How to handle:

Option	When
Remove / Impute	Clearly erroneous values
Keep	Algorithm is tree-based (not sensitive)
Use Robust Scaler	Keep the value, reduce its influence

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06. Feature Engineering

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06. Hand-crafted Features

Creating new features from existing ones using domain knowledge.

Examples:

- bmi | weight / height² | Clinically meaningful
- age_group | bin age into ranges | Captures non-linear patterns
- is_weekend | extract from date | Behavior changes weekends
- revenue_per_user | revenue / users | Normalizes for scale

=> A model can only learn from what you give it. A smart feature can matter more than a sophisticated algorithm.

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07. ML Pipeline

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07. The ML Pipeline



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08. Train/Test Split

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08. Train/Test Split

Split → Scale

The idea: Before training, split your dataset into two parts: one to learn from, one to evaluate on.

Split	Purpose	Typical Size
Training set	Model learns patterns from this	70-80%
Test set	Simulates unseen data	20-30%

=> The idea: Before training, split your dataset into two parts — one to learn from, one to evaluate on. (Studying past exams)

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08. Train/Test Split

Split → Scale

sklearn.model_selection.train_test_split

```
>>> X_train, X_test, y_train, y_test = train_test_split(
... X, y, test_size=0.33, random_state=42)
...
>>> X_train.head()
     sepal length (cm)  sepal width (cm)  petal length (cm)  petal width (cm)
96                 5.7               2.9                4.2               1.3
105                7.6               3.0                6.6               2.1
66                 5.6               3.0                4.5               1.5
0                  5.1               3.5                1.4               0.2
122                7.7               2.8                6.7               2.0
```

08. The three-way split

Split → Scale

Train, Validation & Test Split

Split	Purpose	Typical Size (%)
Training set	Model learns patterns from this	60-70
Validation set	Tune hyperparameters, compare models	15-20
Test set	Final evaluation only (touch once)	15-20

08. Train/Test Split & Leakage

Data leakage: When information from the test set "leaks" into training, making your model look better than it really is.




```
# ❌ WRONG — scaler sees test data
scaler.fit(X)
X_train_scaled = scaler.transform(X_train)
X_test_scaled = scaler.transform(X_test)

# ✅ CORRECT — scaler only fit on train
scaler.fit(X_train)
X_train_scaled = scaler.transform(X_train)
X_test_scaled = scaler.transform(X_test)
```

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Any Questions ?

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Thank you for your attention

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